From Flight Data to Filter Decisions

A corrected, evidence-based comparison of four frozen variometer filters on the first real paragliding flight recording.

Dataset LOG00001 · workbook commit 53907ab · November 2026

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Sources: vario-usb-workbook (commit 53907ab, corrected flight benchmark) · ESP32-S3-Display-DataLogger repository

Companion article: A Flying Breadboard — A Product Development Story

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1 · Executive summary

The first real paragliding flight recording from the project's temporary breadboard logger is of **excellent integrity**: 761.65 s of device time, 30,467 barometer samples at 40.0 Hz and 15,169 vario samples at 19.9 Hz, with zero sequence gaps, duplicates, malformed lines or timestamp discontinuities.

• **The bench pipeline is flight-faithful.** The bench Python C03 reproduces the device-recorded C03 to **3.3 mm/s RMS** — the filter-versus-filter comparison is apples-to-apples.

• **Effective tracking lag is 0.57–0.70 s** on real thermal transitions. This report corrects an erroneous ~2.1 s figure from the first version of the analysis (§5.3).

• All four filters track the flight well; the differences are **matters of degree, not kind**. S4/S5 are measurably quieter (60 vs 67 mm/s RMS; 114 vs 137 mm/s p95) and chatter less (71/80 vs 93/96 audio episodes); C03/D01 have measurably better transient shape after delay compensation (134 vs 197–211 mm/s fast-motion RMS error) and slightly lower overshoot.

• On this dataset the evidence supports **S4 as the airborne default for the next flight**, with C03 retained as the verified reference. This is a measured-evidence-based recommendation, not a universal verdict — §7 states exactly what evidence would reverse it.

2 · Recording setup and sensor configuration

The recording was made by the temporary ESP32-S3 flight logger described in the companion article: a Waveshare ESP32-S3-LCD-1.9 board with a BMP581 barometric breakout, logging to microSD during a paragliding flight. Sensor configuration, as validated during bring-up and unchanged for the flight:

• **BMP581** on the shared I²C bus at 0x47 (chip ID 0x50), oversampling ×16 pressure / ×2 temperature, ODR 40 Hz, on-chip IIR bypass, cadence read; on-chip compensated output Pa = raw24/64.

• One session marker (*S,1*) at file start; boot-persistent sequence counters; record types B (barometer) and V (vario).

• Recording firmware logged the **C03 estimator only** in V frames; D01, S4 and S5 are evaluated by offline replay on the recorded B stream (§4).

• **No IMU data** exists in this recording (§8).

Pressure-to-altitude conversion throughout is the ISA model below, spot-verified against the logger's own altitude field (e.g. raw 5,803,000 -> 90,671.9 Pa -> 927.1 m, matching the recorded 927,204 mm within 8 mm; median offset across the file 7.6 mm).

$$z = 44330\left(1 - (p/101325)^{0.1903}\right)\ [\mathrm{m}]$$

3 · Measurement integrity

Table 1. *Integrity audit of LOG00001 (device-side facts).*

Item	Value
Recording span (device timestamps)	**761.65 s** (12 min 41.6 s)
Barometer (B) rows	**30,467**
Barometer rate	**40.00 Hz** (dt median 24.991 ms, p99 25.106 ms, max 25.421 ms)

Item	Value
Vario (V) rows	15,169
Vario rate	19.92 Hz (dt median 50.004 ms)
B / V sequence integrity	**zero gaps, zero duplicates**
Malformed lines	**0** of 45,637
B timestamps	strictly monotonic; no gaps > 100 ms
Session markers	exactly one (S,1,793567662,1,1) — one continuous session
Filters in V rows	C03 only
IMU rows	none
End of file	normal data rows; last V row 15 ms after last B row — no truncation

The profile is a coherent paragliding sortie (fig. 1): launch and initial climb near t = 0 from ~927 m, sustained thermalling with altitude excursions between 900 and 990 m, and a landing rollout at ~917 m from t ~ 655 s onward — the flat quiet tail used for the noise metrics of §5.

Recording-side observation (no impact on this analysis): the filename LOG00001(2).CSV and non-zero boot counters suggest filename reuse across reboots; whether the SD firmware rotates, appends or overwrites on collision could not be confirmed from this repository. The analyzed file is complete and self-consistent; a filename-allocation audit is queued for the recording-firmware owner.

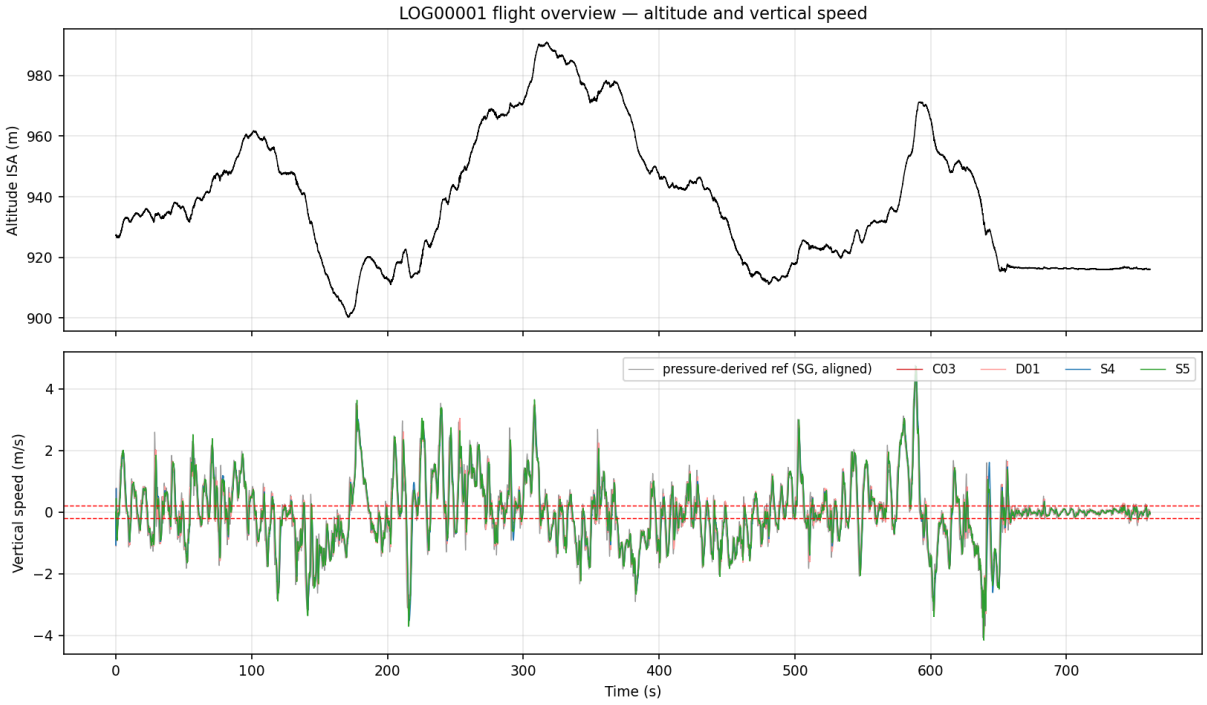


Figure 1. *Full-flight altitude (top) and vertical-speed history with all four filters and the audio thresholds (bottom). Verified plot, reports/flight/fig1_overview.png.*

4 · Methodology: replay, reference, and metric hygiene

Replay. All four filters were replayed on the recorded B stream at real device timestamps (dt from consecutive differences), deterministic causal updates, using the frozen, previously validated Python implementations: **C03** — CV Tracker, q = 1.0, r = 0.035; **D01** — CV Kalman, q = 1.0, r = 0.035; **S4** — fast/slow gated architecture, frozen P4 parameters; **S5** — robust CV Kalman (light gate), frozen P4 parameters. Audio policy: ON at v > +0.20 m/s, OFF at v < +0.05 m/s, applied identically to each replayed output and to the device-recorded C03.

Reference. The pressure-derived reference velocity is a two-stage Savitzky–Golay filter (81-tap on altitude -> gradient -> 41-tap on velocity). The SG window is symmetric about its centre, so the filter is **zero-phase** and the reference is already time-aligned.

Pipeline verification. Replaying C03 and aligning against the device's own V values gives best agreement at -1 B-sample (-25 ms — one 20 Hz V-sample quantization, consistent with causal filtering) with **3.3 mm/s RMS** excluding the first 5 s (shared initialization transient; worst single deviation 1.40 m/s at t = 0.075 s). Audio episodes: 93 replayed vs 94 recorded — agreement within one episode over 12.7 minutes.

Metric hygiene — two rules that keep the numbers honest:

1. **Delay and shape error are measured separately.** Raw cross-correlation lag on detected transitions (|ref| > 0.5 m/s sustained >= 3 s; 46 detected, 44 retained after rejecting windows whose xcorr peak fell on the edge) gives the effective tracking lag; overshoot and settling are then computed against a **delay-compensated** reference so they measure shape, not delay.

2. **Settling/overshoot windows are capped at the next transition start** — thermalling transitions are often only 5–10 s apart, and an uncapped 8 s window would integrate the next thermal burst.

5 · Quantitative comparison

All values measured as described above. “Delay” is the effective cross-correlation lag on real thermal transitions against the aligned pressure-derived reference — a proxy for tracking responsiveness. Treat absolute values with the caveats of §8.

Table 2. *Four-filter comparison on the LOG00001 flight recording (frozen configurations, 40 Hz replay, corrected reference alignment). Bold = best in row.*

Metric	C03	D01	S4	S5
Quiet RMS (mm/s), 37 s coverage	67.2	67.1	**59.8**	**59.5**
Quiet p95 |v| (mm/s)	137.4	137.1	**113.9**	112.9
Transition delay median (s)	**0.600**	**0.575**	0.700	0.625
Transition delay p75 (s)	0.625	0.625	0.750	0.675
Overshoot median, delay-comp (m/s)	**0.277**	0.290	0.349	0.361
Settling median, delay-comp (s)	**7.98**	8.27	9.46	9.71
Fast-motion RMS error, raw (m/s)	0.642	**0.640**	0.732	0.703
Fast-motion RMS error, delay-comp (m/s)	**0.134**	0.145	0.197	0.211
Fast-motion bias (mm/s)	2.2	2.2	-5.9	-5.8

Metric	C03	D01	S4	S5
Output jerk RMS (m/s² per s)	1.034	1.085	**0.894**	0.945
Audio episodes (+0.20 / -0.05)	93	96	**71**	80
Audio ON time (s) / duty	278.5 / 36.6%	278.8 / 36.6%	279.7 / 36.7%	**273.5 / 35.9%**
Mean episode duration (s)	2.99	2.90	**3.94**	3.42
Proxy false triggers (ON >= 2 s while |ref| < 0.15)	0	0	0	0

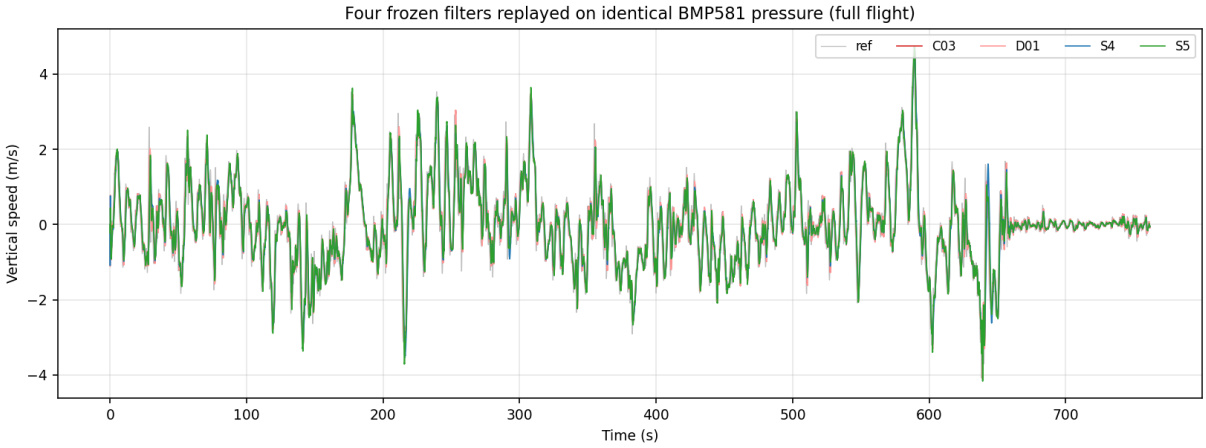


Figure 2. *All four filters replayed on identical BMP581 pressure input, full flight. The outputs are nearly coincident — the differences are matters of degree. Verified plot, reports/flight/fig2_overlay.png.*

5.1 · Reading the table

Quietness (fig. 5): S4/S5 are ~16% lower in RMS and ~18% lower in p95 than C03/D01 in the quiet segments — the largest robust difference in the study. But quiet coverage is 37 s of landing rollout only; there is no long quiet-soaring stretch in this flight.

Transient shape (fig. 3): C03/D01 win the delay-compensated fast-motion error (134–145 vs 197–211 mm/s), overshoot (0.28–0.29 vs 0.35–0.36 m/s) and settling (8.0–8.3 vs 9.5–9.7 s).

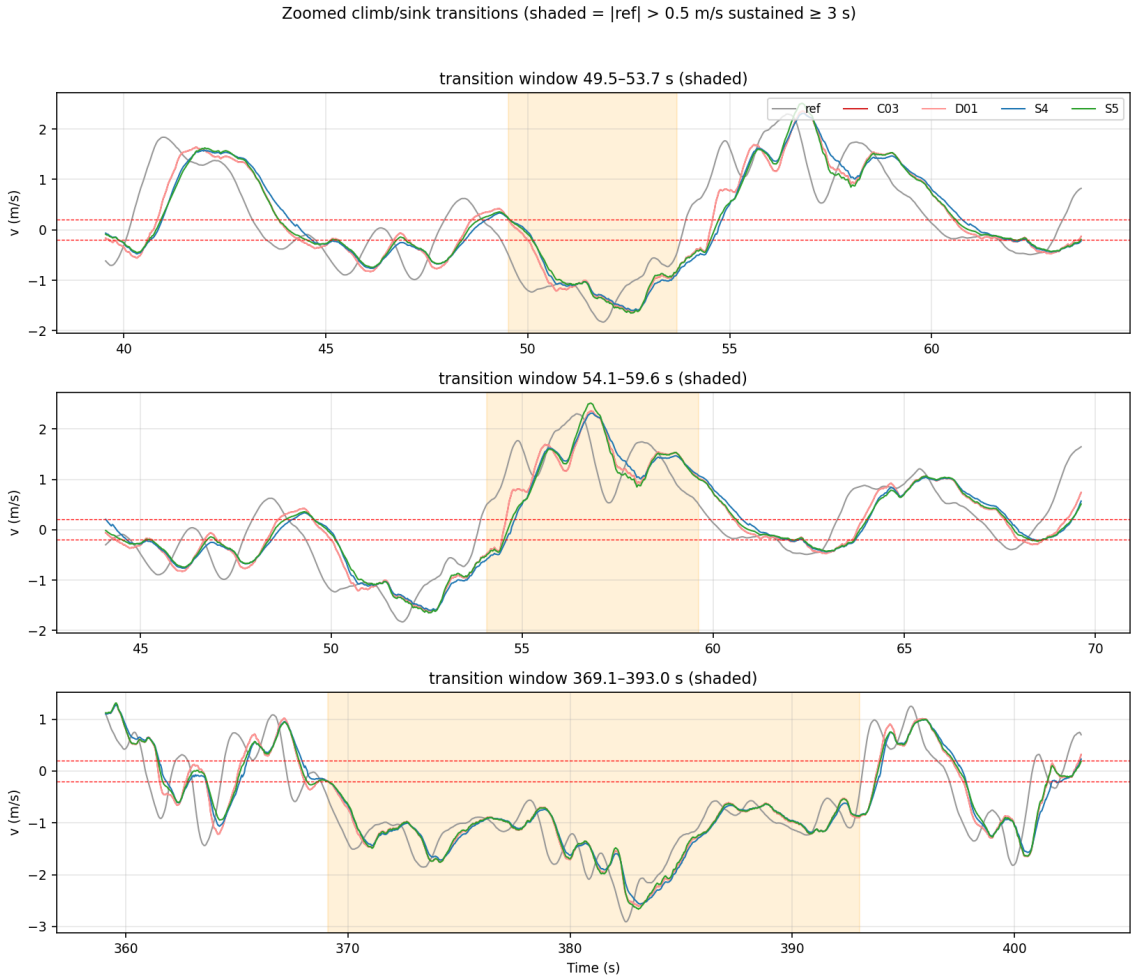


Figure 3. *Three zoomed climb/sink transition windows (shaded), showing onset, tracking and exit behaviour per filter against the aligned reference. Verified plot, reports/flight/fig3_transitions.png.*

Chatter (fig. 4): total audio ON-time is nearly identical across filters (274–280 s of 761 s ~ 36% duty) — all four capture essentially the same lift. The difference is how that time is broken up: C03/D01 produce 93–96 short episodes, S4 71 longer ones, S5 80. With zero proxy false triggers for every filter, the extra episodes are mostly re-entries around the +0.20 threshold in turbulent, near-threshold air — not demonstrably wrong, but audible.

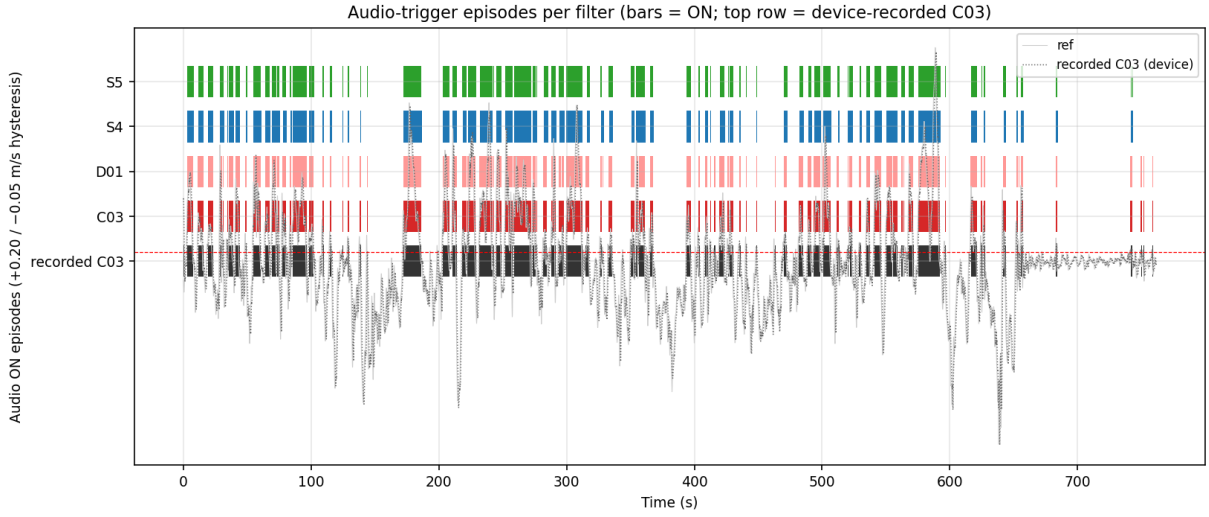


Figure 4. *Audio ON-episode timelines per filter versus device-recorded C03, over the reference velocity. Episode patterns track the same thermal bands; S4/S5 group them into fewer, longer episodes. Verified plot, reports/flight/fig5_audio.png.*

Delay: all four respond in ~0.6–0.7 s. C03/D01 are marginally faster (0.58–0.60 s) than S5 (0.63 s) and S4 (0.70 s). Proportionally the spread is larger than it first appears (S4 ~17% slower than D01), but in absolute terms all four sit in the same responsiveness class.

Smoothness: S4 has the lowest output jerk (0.894 vs 1.03–1.09), consistent with its gating architecture suppressing small-scale velocity jitter. C03 and D01 are nearly identical on every metric, as expected from their shared q/r heritage — the practical field on this flight is {C03/D01} vs {S4, S5}.

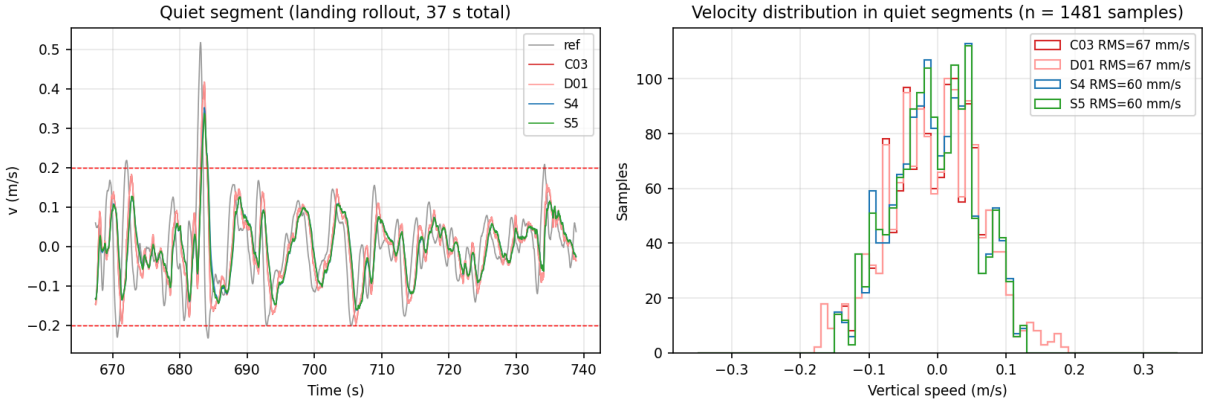


Figure 5. *Left: landing-rollout quiet segment (37 s total). Right: per-filter velocity-distribution histograms in the quiet segments — the measurable basis for the quietness ranking. Verified plot, reports/flight/fig4_quiet.png.*

The responsiveness–smoothness trade-off in one line: C03/D01 give crisper transients; S4/S5 give quieter output and calmer audio with slightly softer transient fidelity. Neither class is wrong; they weight the same measured trade differently.

5.2 · Corrected response-lag methodology — why ~2.1 s was wrong

The first published version of this analysis reported an effective lag of ~2.1 s. **That value was invalid**, for a subtle and instructive reason.

The pressure-derived reference uses a **centred (symmetric) Savitzky–Golay filter**. A symmetric SG filter is zero-phase: it has no group delay to correct. The erroneous run applied a “group-delay correction” of SG_LAG = 60 samples — a rule that holds for *causal/trailing* windows, not for a centred one — shifting the reference 1.5 s **beyond** truth. Every delay-conflated metric inherited exactly that 1.5 s inflation: filters appeared to lag ~2.1 s when the true lag was ~0.6 s, a discrepancy confirmed independently by cross-correlation.

This revision removes the shift (the corrected script carries a comment recording the mistake so it is not reintroduced). The shape-only metrics — delay-compensated overshoot, settling and fast-motion error — were essentially unaffected, as they should be. **The corrected lag of 0.57–0.70 s is a pressure-derived, internally consistent estimate of effective tracking responsiveness.** It is not an independently verified physical response time: with no independent vertical-speed ground truth (§8), it should be read as “how far each filter trails the best pressure-only estimate of truth”, not as truth itself.

5.3 · Audio threshold behaviour

The +0.20 / -0.05 m/s hysteresis policy behaves on flight data exactly as designed in the virtual-audio bench work. In turbulent air near threshold, output hovering around +0.20 m/s produces clusters of short re-entry episodes (C03/D01) versus fewer, longer sustained episodes (S4's gating extends mean episode length to 3.9 s). Episode patterns track the same thermal bands across all four filters, total ON-time is equal, and the proxy false-trigger screen (ON >= 2 s while the reference is below 0.15 m/s in magnitude) finds **zero** violations for every filter. The practical interpretation: the threshold itself is not generating false lift indications on this flight; the remaining design lever is how calmly the threshold is approached — a filter property, not a policy property.

6 · Ranking on this dataset, and the recommended default

Table 3. *Real-world paragliding ranking — conditional on this single flight.*

Rank	Filter	Basis
1	S4	Quietest-class noise, fewest audio episodes, lowest jerk, zero proxy false triggers; costs are real but bounded (+0.1 s onset delay, +0.07 m/s median overshoot, +0.06–0.13 m/s delay-compensated fast-motion error). Best overall pilot-experience metrics.
2	S5	Statistically tied with S4 on quietness; middling chatter; better fast-motion shape than S4 but worse than C03/D01.
3	C03	Best transient fidelity and fastest settling; verified identical to device firmware; noticeably chattier audio and noisier quiet output.
4	D01	Same behaviour as C03 within measurement noise; no measured advantage over C03 on this data.

Recommended default for the next flight: S4, with C03 retained as the verified reference implementation. A pilot prioritizing crisp lift onset over audio calm would rationally pick C03 instead — both choices are defensible on this data.

What would reverse this recommendation: (a) a flight with long quiet soaring stretches confirming or refuting that the quiet advantage generalizes beyond landing rollout; or (b) GPS reference data showing S4's gating misses or delays weak-but-flyable lift onsets. Either result feeds back before anything is frozen — this remains a design-space investigation, not a selection.

7 · Limitations

• **No independent ground truth.** No GPS, video or pilot-log timestamps. All response/error figures use the pressure-derived SG reference — internally consistent proxies, not truth. The corrected lag in particular must not be cited as an independently verified physical response time.

• **Single flight.** One air mass, one pilot, one day. Rankings are conditional on this dataset.

• **No IMU data.** FUS (baro+IMU fusion) is unevaluable on flight data; no attitude/acceleration context exists for interpreting filter behaviour; all fusion gains from bench tests remain unvalidated in flight.

• **Limited quiet coverage.** The quiet metrics rest on 37 s of landing rollout; there is no long soaring-quiet stretch in this flight.

• **Launch/landing times are inferred** from the data, not logged.

• **Replay scope.** D01/S4/S5 were evaluated by offline replay; the recording firmware logged C03 only. On-target CPU/timing for the other estimators stands from previous bench phases.

• **Recording-firmware filename behaviour unverified** (§3 observation).

8 · Next validation experiments

• **Log multiple estimators to SD** — interleaved V frames for D01/S4/S5 remove the replay-verification step entirely and give on-device comparison.

• **Add IMU + GPS** to flight recordings — enables FUS evaluation, an independent velocity reference, and launch/landing annotation.

• **A quiet-soaring flight** — 20+ minutes of straight-glide/soaring to firm up the quietness ranking beyond rollout data.

• **Repeat with S4 airborne** to test the §6 recommendation against weak-lift onsets.

• **SD filename-rotation audit** of the recording firmware (§3).

Reproducibility

• Dataset copy: analysis/flight/LOG00001_flight.csv

• Analysis: analysis/flight/flight_analysis.py -> flight_metrics.json + replay_series.npz; figures: analysis/flight/flight_plots.py -> reports/flight/

• Filter implementations: filter_benchmark/src/filters/ (frozen, previously validated); workbook revision: commit 53907ab (corrected reference alignment)

• Recording hardware and flight context: companion article *A Flying Breadboard* and the ESP32-S3-Display-DataLogger repository.